How Much Room Is Left Below PPI++? Efficiency Limits and Cross-Fitted Recalibration for Prediction-Powered Mean Inference When Labels, Unlabeled Data and Predictor Are Fixed

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Abstract

Prediction-powered inference (PPI) builds valid confidence intervals from a small labeled sample, a large unlabeled sample and a fixed pre-trained predictor. We ask a narrow question: with all three held fixed and nominal coverage required, how much shorter than PPI/PPI++ can an interval for the population mean be, and when? We show that, if the predictor output is the only information used, the answer is exactly the non-affine part of the calibration curve $\mathbb{E}[Y \mid f(X)]$: PPI++ is asymptotically efficient if and only if this curve is affine (or there is no unlabeled data), and the width reduction available to any estimator is at most $1 - \sqrt{1 - sN/(n+N)}$, where s is the share of residual variance that is non-affine. A cross-fitted recalibrated estimator (isotonic, spline or CV-selected) attains the bound and stays valid for any calibration fit, without convergence rates. In Monte-Carlo experiments with known truth (2000 replicates per synthetic cell, 1000 per real-data cell), on seven example data sets from `ppi-python` the attainable gain over PPI++ is at most 1.2% in width (nonlinear share $\leq 2.5\%$) and the recalibrated intervals realise essentially none of it: at $n = 500$ they are 0.5–2% wider than PPI++ on six sets and 0.6% narrower on one, and up to 19% wider at $n = 50$. Gains appear when the predictor outputs are not on the scale of the label (logit or otherwise monotonically distorted scores): 2–27% shorter on distorted versions of the same data, within about 2 percentage points of the bound. A hard binary predictor offers no headroom, and under distribution shift the recalibrated intervals lose coverage completely where PPI keeps it. Separately, ppi_py's clipping of the PPI++ weight costs about 5% width on one data set. Our conclusion is conditional: PPI++ is close to optimal for well-calibrated predictors, and recalibration is worth it only for predictors whose scores are visibly non-affine in the label.

1 Introduction

In ecology, astronomy and social science a typical analysis has three ingredients: a small set of n records with expensive gold-standard labels, a much larger set of N records with features only, and a pre-trained model that predicts labels for any record. Prediction-powered inference (PPI) [Angelopoulos et al., 2023a] and its power-tuned version PPI++ [Angelopoulos et al., 2023b] use the predictions to shrink confidence intervals while keeping nominal coverage, regardless of how good the predictor is. Subsequent work changes the inputs: it trains the predictor on the labels with cross-fitting [Zrnic and Candès, 2023], spends the label budget more cleverly [Zrnic and Candès, 2024], or stratifies the labeled set [Fisch et al., 2024]. Here we fix everything: the labeled set, the unlabeled set and the predictor are given, and nothing can be added.

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Question. *Given exactly these inputs, can the interval be made shorter than PPI/PPI++ without losing coverage; by how much; and under which conditions?*

We study the population mean $\theta = \mathbb{E}[Y]$. It is the simplest target for which a sharp answer is available and a closed-form efficiency analysis is possible; it is also the target of every example in the `ppi-python` data used here. Our answer has three parts.

1. **A tight characterization (Sec. 3).** If the predictor output $Z = f(X)$ is the only information used, the asymptotic variance of any regular estimator is at least $\mathbb{E}\mathrm{Var}(Y|Z)/n + \mathrm{Var}(\mathbb{E}[Y|Z])/(n+N)$. PPI++ attains it iff $\mathbb{E}[Y|Z]$ is affine in Z or $N/(n+N) \to 0$; otherwise the variance gap is $\mathrm{Var}(h)\, N/(n(n+N))$ with h the non-affine part of the calibration curve. A hard 0/1 predictor can never be improved. This is the semiparametric theory of missing-at-random means [Robins et al., 1994, Tsiatis, 2006] applied to PPI; it is also close to what is known for semi-supervised mean estimation [Zhang et al., 2019], so we do not claim novelty for the bound itself but for its use as a yardstick for PPI in the fixed-inputs regime.
2. **A route to the bound (Sec. 4).** A cross-fitted recalibrated PPI interval whose validity needs only L^2-stability of the calibration fit, not a rate, and that is therefore valid for *any* calibration learner (isotonic, spline, bins, CV-selected).
3. **Evidence about when it pays (Sec. 5).** Across synthetic data with known nonlinearity and seven real data sets resampled with known truth, we measure coverage and width with uncertainty, and compare with the closed-form bound. The headline is negative for the common case and positive for a specific, identifiable case: gains require a predictor whose scores are non-affine in the label; on the unmodified predictors that ship with `ppi-python` there is essentially nothing left to gain from the predictor output.

2 Related work

Prediction-powered inference. PPI [Angelopoulos et al., 2023a] gives valid intervals for means, quantiles and regression coefficients from a labeled set, an unlabeled set and a black-box predictor; PPI++ [Angelopoulos et al., 2023b] adds a data-driven weight λ that guarantees (asymptotically) no loss relative to the classical interval. Cross-PPI [Zrnic and Candès, 2023] handles predictors trained on the same labels; active inference [Zrnic and Candès, 2024] chooses which points to label; stratified PPI [Fisch et al., 2024] allocates labels across strata of the predictor. All of these change a part of the input or the sampling design. Our setting excludes that; the closest ingredient is stratification on the predictor, which we include as a post-stratification baseline (bin10). Miao et al. [2023] propose assumption-lean, data-adaptive post-prediction inference, another way of using ML predictions for valid inference with a small labeled set.

Semi-supervised and missing-data inference. Estimating a mean from a labeled and an unlabeled sample has a classical efficiency theory [Robins et al., 1994, Bickel et al., 1993, Tsiatis, 2006], developed for the semi-supervised mean in Zhang et al. [2019], with high-dimensional covariates in Zhang and Bradic [2022] and M-estimation in Song et al. [2024]. PPI++ for the mean is a member of the class of estimators that are linear in the predictor; what has not, to our knowledge, been laid out is how large the gap to the bound is for the predictors and data sets on which PPI is demonstrated, or its closed form in PPI's notation. Survey statistics has long used nonparametric model-calibrated estimators with the same structure [Särndal et al., 1992, Montanari and Ranalli, 2005], and double machine learning provides the cross-fitting device [Chernozhukov et al., 2018].

Calibration of classifiers. That predicted scores need not be on the scale of the label is well known: boosted trees and similar models produce sigmoid-shaped distortions of probabilities [Niculescu-Mizil and Caruana, 2005] and modern networks are over-confident [Guo et al., 2017]. This is exactly the situation in which a linear weight λ cannot correct the predictor.

3 Setting and what is attainable when everything is fixed

Setting. Let (X_i, Y_i), $i = 1, \ldots, m$, be i.i.d. with $m = n + N$. A uniformly random subset L of size n is labeled; for $i \notin L$ the label is missing (missing completely at random, MCAR, by design of the labeling). The pre-trained predictor f is fixed and not trained on any of these data; write

$Z_i = f(X_i) \in \mathbb{R}$ (the argument below only uses that Z_i is a measurable function of X_i). The analyst sees Z_i for all m points and Y_i for $i \in L$, and wants a $(1-\alpha)$ interval for $\theta = \mathbb{E}[Y]$. We chose the mean because it is the simplest target for which the efficiency question has a closed-form answer, it is the target for which PPI was first demonstrated, and it is what all nine example data sets in `ppi-python` are used for. Write

$$g(z) = \mathbb{E}[Y \mid Z = z], \quad \sigma^2 = \mathbb{E}\,\mathrm{Var}(Y \mid Z), \quad V = \mathrm{Var}(Z), \quad \beta = \tfrac{\mathrm{Cov}(Y,Z)}{V}, \quad h(z) = g(z) - \mathbb{E}g - \beta(z - \mathbb{E}Z),$$

so h is the part of the calibration curve g that is *not* affine in the predictor output, and $v_h = \mathrm{Var}\, h(Z)$. Let $s = v_h/(\sigma^2 + v_h)$ be the *nonlinear share* of the residual variance left by the best affine recalibration. All variances below are asymptotic variances of the estimator of θ (i.e. squared interval half-widths up to the factor $z^2_{1-\alpha/2}$); $n, N \to \infty$ with $n/m \to \pi \in (0, 1]$.

Existing estimators. PPI with weight λ uses $\hat\theta_\lambda = \frac{1}{N}\sum_{j \notin L} \lambda Z_j + \frac{1}{n}\sum_{i \in L}(Y_i - \lambda Z_i)$ [Angelopoulos et al., 2023a,b]; $\lambda = 1$ is the original PPI, $\lambda = 0$ the classical sample mean, and PPI++ picks λ from the data to minimize variance. Its asymptotic variance is $\mathrm{Var}(Y - \lambda Z)/n + \lambda^2 V/N$.

Proposition 1 (Variance of the PPI family)**.** *Minimizing over $\lambda \in \mathbb{R}$ gives $\lambda^\star = \beta N/m$ and $V_{\mathrm{PPI++}} = \dfrac{\sigma^2 + v_h}{n} + \dfrac{\beta^2 V}{m}$. For fixed λ, $V_\lambda - V_{\mathrm{PPI++}} = V\frac{m}{nN}(\lambda - \lambda^\star)^2$; in particular PPI with $\lambda = 1$ is suboptimal by $V\frac{m}{nN}(1 - \beta N/m)^2$, which vanishes only if $\beta N/m = 1$.*

Proposition 2 (Efficiency bound with the predictor output as the only information)**.** *In the model where (Z, Y) has an unrestricted distribution (finite variance), any regular asymptotically linear estimator of θ based on $\{Z_i\}_{i \le m}$ and $\{Y_i\}_{i \in L}$ has asymptotic variance at least*

$$V_{\mathrm{eff}} = \frac{\sigma^2}{n} + \frac{\mathrm{Var}\, g(Z)}{m} = \frac{\sigma^2}{n} + \frac{\beta^2 V + v_h}{m}.$$

It is attained by $\hat\theta = \frac{1}{n}\sum_L (Y_i - g(Z_i)) + \frac{1}{m}\sum_{i \le m} g(Z_i)$, whose influence function is $\frac{R}{\pi}(Y - g(Z)) + g(Z) - \theta$.

Proof sketch. This is the standard semiparametric result for the mean under MCAR missingness with known labeling probability π [Robins et al., 1994, Bickel et al., 1993, Tsiatis, 2006]: with the nonparametric model for (Z, Y), the observed-data influence function of θ is unique and equals $\frac{R}{\pi}(Y - \theta) - \frac{R - \pi}{\pi}\mathbb{E}[Y - \theta \mid Z]$. Its variance per observation is $\mathbb{E}(Y - g)^2/\pi + \mathrm{Var}\, g$, which in units of the sample sizes is the displayed expression. The case $Z = \text{const}$ recovers $\mathrm{Var}(Y)/n$, and $N = \infty$ recovers σ^2/n [Zhang et al., 2019]. □

Corollary 1 (When PPI++ cannot be improved)**.** $V_{\mathrm{PPI++}} - V_{\mathrm{eff}} = v_h \dfrac{N}{n\,m}$, *hence*

$$\frac{V_{\mathrm{eff}}}{V_{\mathrm{PPI++}}} = 1 - \frac{N}{m} \cdot \frac{s}{1 + \kappa} \;\ge\; 1 - \frac{N}{m}\, s, \qquad \kappa = \frac{n}{m}\frac{\beta^2 V}{\sigma^2 + v_h} \ge 0.$$

Thus (i) PPI++ is asymptotically efficient if and only if g is a.s. affine in Z ($v_h = 0$) or $N/m \to 0$ (essentially no unlabeled data); (ii) whenever the predictor output takes two values (a hard 0/1 prediction), g is automatically affine, so no estimator that uses only Z can beat PPI++; (iii) the interval-width reduction is at most $1 - \sqrt{1 - sN/m} \le 1 - \sqrt{1 - s}$, e.g. about 5% for $s = 0.10$ and about 1% for $s = 0.02$.

The corollary follows by subtracting the two expressions in Props. 1 and 2. It says exactly *what* room remains below PPI++ with labeled set, unlabeled set and predictor all fixed: only the non-affine part of the calibration curve $\mathbb{E}[Y \mid f]$, discounted by the fraction N/m of unlabeled data. It also shows that the large-N regime that motivates PPI is where the headroom is largest. Using side features X beyond $Z = f(X)$ lowers the bound further to $\mathbb{E}\,\mathrm{Var}(Y|X)/n + \mathrm{Var}(\mathbb{E}[Y|X])/m \le V_{\mathrm{eff}}$, but attaining it requires learning $\mathbb{E}[Y|X]$ from the labels, i.e. in effect replacing the predictor; we only quantify how large this extra gap is on the data at hand (§5.5).

Remark 1 (Invariance). A monotone reparametrization $Z \mapsto T(Z)$ (e.g. logits instead of probabilities) leaves g-based estimators with monotone or nonparametric g unchanged, but changes β, v_h and hence the width of PPI/PPI++. A predictor whose scores are not on the scale of Y is therefore exactly the case with $v_h > 0$.

4 Cross-fitted recalibrated PPI

Proposition 2 suggests the obvious route to shorter intervals: estimate the calibration curve $g = \mathbb{E}[Y|Z]$ from the labeled data instead of fixing it to $z \mapsto \lambda z$. The estimator is an augmented-inverse-probability-weighted estimator with known labeling probability [Robins et al., 1994], equivalently a model-assisted/model-calibrated survey estimator with a nonparametric working model [Särndal et al., 1992, Montanari and Ranalli, 2005]; it is also the mean-estimation special case of cross-fitted DML [Chernozhukov et al., 2018]. We do not claim the estimator is new; the point of this paper is to measure how much it can gain over PPI/PPI++ and when.

Algorithm. Split the labeled indices into $K = 5$ random folds $L_1, \dots, L_K$. For each k fit $\hat{g}_{-k}$ on $L \setminus L_k$ by a one-dimensional regression of Y on Z. Put $\hat{g}_i = \hat{g}_{-k(i)}(Z_i)$ for $i \in L$ (out-of-fold) and $\bar{g}_j = \frac{1}{K}\sum_k \hat{g}_{-k}(Z_j)$ for unlabeled j. Then

$$\hat{\theta} = \frac{1}{n}\sum_{i\in L}(Y_i - \hat{g}_i) + \frac{1}{m}\Big(\sum_{i\in L}\hat{g}_i + \sum_{j\notin L}\bar{g}_j\Big), \qquad \hat{V} = \frac{\widehat{\mathrm{Var}}_L(Y-\hat{g})}{n} + \frac{\widehat{\mathrm{Var}}_{\mathrm{all}}(\hat{g})}{m},$$

and the interval is $\hat{\theta} \pm z_{1-\alpha/2}\sqrt{\hat{V}}$ (sample variances with $n-1$ and $m-1$ denominators). With $\hat{g} \equiv \lambda z$ this is PPI up to the use of the labeled predictions in the second mean; with $\hat{g}$ the OLS line it is the survey regression estimator, asymptotically equivalent to power-tuned PPI. Learners (all in `code/cppi.py`): **lin** (OLS line); **iso** (isotonic regression, increasing, Barlow et al., 1972); **bin10** (10 equal-mass bins, i.e. post-stratification on the predictor output, in the spirit of stratified PPI Fisch et al., 2024 but with random rather than allocated labels); **spl** (natural cubic regression spline with 5 knots at the 5%–95% quantiles of the training scores, Hastie et al., 2009); **auto** (choose among lin, iso, spl by inner 4-fold CV mean squared error, nested inside each outer fold, so that selection never sees the points it is evaluated on). No tuning on the evaluation data was done; K, knot counts and bin counts were fixed before the experiments.

Proposition 3 (Validity)**.** *Suppose $\mathbb{E}Y^2 < \infty$ and there is a function g^* with $\mathbb{E}g^*(Z)^2 < \infty$ such that $\mathbb{E}[(\hat{g}_{-k}(Z) - g^*(Z))^2 \mid \text{training fold}] \to_p 0$ for each k, where Z is an independent copy. Then $(\hat{\theta} - \theta)/\sqrt{\hat{V}} \Rightarrow N(0,1)$, so the interval has asymptotic coverage $1-\alpha$. Moreover $\hat{\theta}$ is exactly conditionally unbiased given the training folds for* any *$\hat{g}$. If $g^* = g$ the interval attains V_{eff}.*

The limit g^* need not be the truth: a misspecified learner (e.g. a monotone fit to a non-monotone g) yields a valid but longer interval, with variance $\mathbb{E}(Y - g^*)^2/n + \mathrm{Var}\, g^*(Z)/m$. This is the same robustness-to-the-predictor that makes PPI valid for any f; the proof (Appendix A) uses only conditional Chebyshev bounds and no convergence *rate* for $\hat{g}$. The guarantee is asymptotic; Sec. 5 measures finite-sample coverage down to $n = 25$. Data-driven learner selection (**auto**) is covered by the proposition when the selected learner's limit is stable; we do not prove this and rely on simulation.

5 Experiments

Protocol. Every number comes from Monte-Carlo replicates with a known truth. *Real-data cells:* the seven `ppi-python` example sets that contain $(Y, \hat{Y})$ for all records (Galaxies P=16,743, Forest 1,596, AlphaFold 10,802, Gene expression 61,150, Census income 380,091, Census health insurance 318,215, Census education 318,215 records; P is the population size) are each treated as the population; a replicate draws n labeled and N unlabeled records i.i.d. *with replacement* and the truth is the population mean of Y. The predictor is the one shipped with the data, never refit. (Records repeat when $n + N$ is not small relative to P, most visibly for Forest; the DGP is still i.i.d. from the empirical distribution, but the efficiency bound of Prop. 2 for that DGP is then not the one we compute from cross-validated fits, so we flag Forest at $n \geq 1000$.) *Synthetic cells:* (a) $f \sim N(0,1)$, $Y = f + c\,(f^2 - 1)/\sqrt{2} + \varepsilon$, $\varepsilon \sim N(0,1)$, so $s = c^2/(1+c^2)$ is known exactly; (b) a classifier that outputs logits $f \sim N(\mu, \tau^2)$ with $Y \sim \mathrm{Bernoulli}(\mathrm{expit}(f))$ (so $\mathbb{E}[Y|f] = \mathrm{expit}(f)$ is non-affine), with $(\mu, \tau) = (-1, 2)$ ($s = 0.06$) and $(-1, 4)$ ($s = 0.23$). *Methods:* classical; PPI (λ=1) and PPI++ from `ppi-python` 0.2.3 (which clips $\hat{\lambda}$ to $[0,1]$); PPI++ with the unclipped plug-in $\hat{\lambda} = \widehat{\mathrm{Cov}}(Y,f)/((1+n/N)\widehat{\mathrm{Var}}(f))$, which is the best member of the PPI family and is our reference;

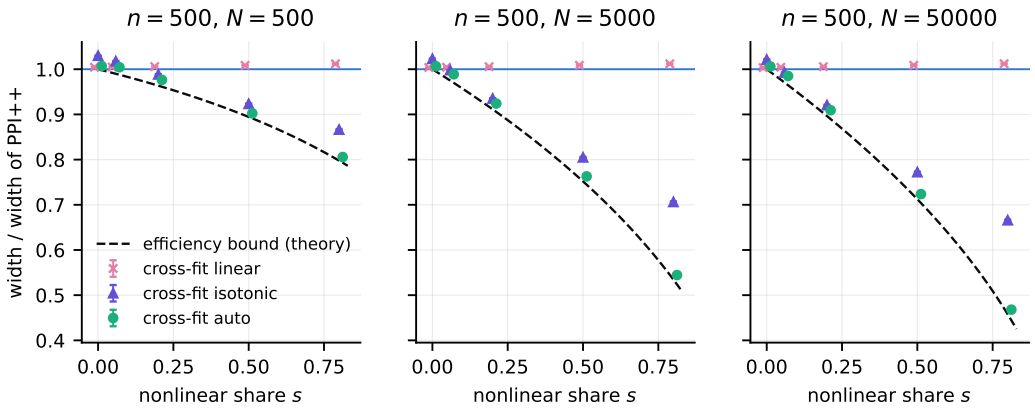


Figure 1: Width of cross-fitted intervals relative to PPI++ (unclipped λ) versus nonlinear share s of the calibration curve ($n = 500$; synthetic quadratic DGP; 2000 replicates; bars: ± 1.96 s.e.). Dashed: closed-form bound $\sqrt{V_{\text{eff}}/V_{\text{PPI++}}}$.

and cross-fitted lin / bin10 / iso / spl / auto (Sec. 4, $K = 5$). All intervals are 95% Wald-type. *Metrics:* coverage with Wilson intervals; width reported as the paired per-replicate ratio to the reference (mean ± 1.96 s.e. over replicates, so values that differ from 1 by more than the $\pm$ are not noise); the closed-form ratio $\sqrt{V_{\text{eff}}/V_{\text{PPI++}}}$ from Props. 1–2 ("bound") computed from the population (real data: 5-fold cross-validated isotonic/spline fit of $\mathbb{E}[Y|f]$, so it is itself an estimate, and is capped by the linear fit). Replicates: 2000 per synthetic cell, 1000 per real-data cell. Seeds are derived from (experiment cell, replicate index), so the same labeled/unlabeled draws are shared across methods within a cell.

5.1 Synthetic data: the bound is tight, and recalibration reaches it

Figure 1 varies the nonlinear share s at $n = 500$. The cross-fitted linear interval tracks PPI++ (within 1.2%, the small price of cross-fitting), while spline/auto intervals follow the efficiency bound: for N=50,000 the width ratio to PPI++ is 0.47 at $s = 0.8$, 0.72 at $s = 0.5$, 0.91 at $s = 0.2$ against the bound 0.46, 0.71, 0.90. Over the 20 cells the empirical ratio of spline/auto exceeds the bound by 0.010 ± 0.003 (mean $\pm$ s.d. over cells; maximum 0.015), which is the cost of estimating the calibration curve. Gains vanish when N is small (N=100: at most 5.5% at $s = 0.8$), as Cor. 1 predicts. Isotonic regression falls short here because $c(f^2 - 1)$ is non-monotone: its limit $g^* \neq g$, so the interval stays valid but longer (0.67 vs 0.47), as Prop. 3 allows. Coverage of auto was between 94.2% and 95.9% in all 20 cells (the two cells where several methods sit at 93.4–94.0% are shared by PPI and PPI++ because the replicates are shared; Table 4).

Small samples (Fig. 2). The price of flexibility is paid at small n. With a linear truth ($c = 0$), auto is 14.5%, 7.1%, 3.4%, 0.6% wider than PPI++ at $n = 25, 50, 100, 500$; spline alone is 65% wider at $n = 25$. With a quadratic truth ($c = 1$) auto breaks even at about $n = 30$ and gains 10.1% ($n = 50$), 20% ($n = 100$), 27% ($n = 500$). Wald coverage at $n = 25$ is 91–95% across methods (classical 92.7–93.0%); the cross-fitted intervals are not worse and, at $n = 25$–50, better than ppi_py's PPI++ (e.g. $c = 1$, $n = 50$: 93.2% auto vs 91.8% PPI++, 92.0% for the t-version of unclipped PPI++ vs 93.6% for auto). A Student-t calibration of the half-widths with $n - 1$ d.f. narrows but does not close this gap for PPI++ (92.0% at $n = 50$).

5.2 Real data with the shipped predictors: essentially nothing to gain

Table 1 (and Appendix Figs./Tables) shows the central negative result. The nonlinear share of the calibration curve is at most 2.5% (Gene expression) and $\leq 0.5\%$ for the other six data sets; the implied bound on the achievable width reduction over PPI++ at $n = 500$, $N = 20{,}000$ is 0.9% (Gene expression), 0.2% (Census income) and $\leq 0.1\%$ elsewhere. The cross-fitted auto interval does not realise even that: at $n = 500$ it is 0.5–2.1% *wider* than PPI++ on six data sets, and 0.6%

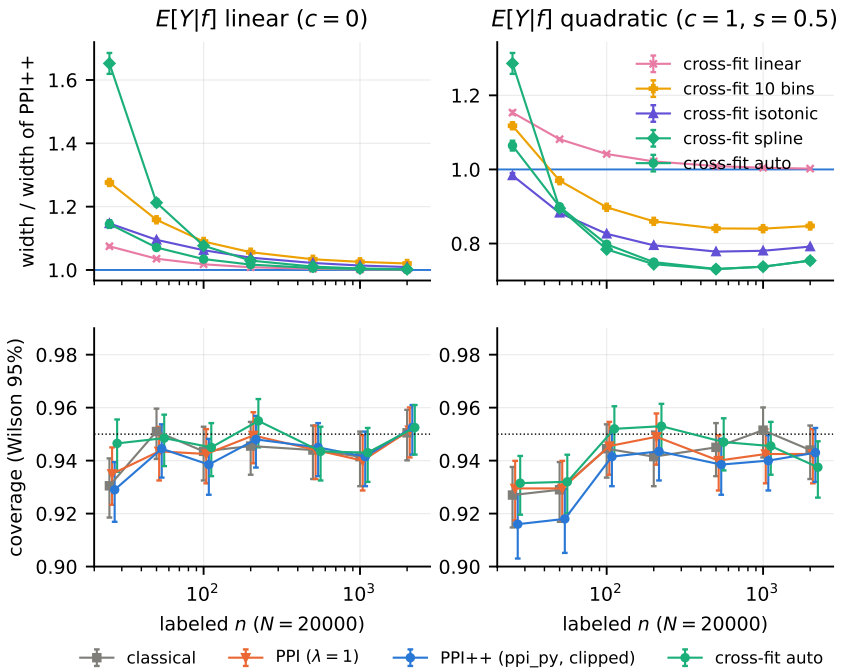


Figure 2: Small-n behaviour ($N = 20{,}000$, quadratic synthetic DGP, 2000 replicates). Top: width relative to PPI++ (unclipped). Bottom: coverage with Wilson 95% intervals.

Table 1: Real data, shipped predictors, $n = 500$, $N = 20{,}000$, 1000 replicates. corr: correlation of Y and $\hat{Y}$; $\hat{\beta}$ slope of Y on $\hat{Y}$; s: cross-validated nonlinear share of residual variance; widths are ratios to PPI++ with unclipped $\hat{\lambda}$ ($\pm$: 1.96 s.e.); "bound" is $\sqrt{V_{\text{eff}}/V_{\text{PPI++}}}$ estimated from the population. Forest has only 1596 records ($n/P = 0.31$).

				coverage (%)				width / PPI++ (unclipped)			
data set	corr	$\hat{\beta}$	s	classical	PPI	PPI++	auto	classical	PPI	auto	bound
Galaxies	0.74	0.98	0.0%	95.3	95.1	94.7	95.2	1.47	1.00	1.007 $\pm$0.000	1.000
Forest	0.54	0.90	0.0%	96.1	95.3	95.1	95.1	1.18	1.00	1.005 $\pm$0.001	1.000
AlphaFold	0.82	1.03	0.2%	95.0	94.8	94.3	94.2	1.69	0.99	1.005 $\pm$0.001	0.999
Gene expression	0.96	1.15	2.5%	94.1	93.9	94.0	93.8	3.24	1.05	0.994 $\pm$0.001	0.991
Census income	0.61	1.38	0.5%	94.2	93.5	93.1	93.4	1.26	1.02	1.021 $\pm$0.002	0.998
Census health ins.	0.71	0.99	0.1%	94.2	94.7	94.8	95.0	1.40	1.00	1.008 $\pm$0.000	1.000
Census education	0.63	0.98	0.0%	93.9	94.7	94.1	94.5	1.28	1.00	1.011 $\pm$0.001	1.000

narrower on Gene expression; at $n = 2000$ all six are within 0.6% of PPI++ (the 1% "gains" on Forest at $n \geq 1000$ are the resampling artifact flagged above); at $n = 50$ auto is 3–19% wider except on Gene expression (-2.4%). At $n \geq 500$ the coverage of every method lies between 91.5% and 96.8% (mean 94.5–95.2%); the lowest cells (92.2–92.8%, Census income with $N = 5000$) are shared by all methods including the classical interval. At smaller n the Census-income coverage is 82–92% for $n \leq 200$ for every method (heavy-tailed income), so these shortfalls come from the Wald approximation and not from recalibration.

Two other facts matter for practice. (i) The headroom *relative to the classical interval* is large (width 1.2–3.2 times higher), so using a predictor is worthwhile; the headroom beyond PPI++ is not. (ii) The choice *within* the PPI family matters more than recalibration: PPI with $\lambda = 1$ is 4.9% wider than PPI++ on Gene expression ($\hat{\beta} = 1.15$) and 1.8% on Census income ($\hat{\beta} = 1.38$), and ppi_py's PPI++ shows the same loss because its $\hat{\lambda}$ is clipped at 1 while the unclipped optimum is above 1 (bound ratio of the clipped version to the unclipped one: 1.056 and 1.018).

Table 2: Monotonically distorted predictor outputs, $n = 1000$, $N = 20{,}000$, 1000 replicates. Width ratios to PPI++ with unclipped $\hat{\lambda}$. The last column is the ratio of ppi_py's clipped PPI++ to the unclipped one.

		width / PPI++ (unclipped)				coverage (%)			
predictor output	s	PPI	iso	auto	bound	PPI++	iso	auto	λ^{clip}-PPI++
Galaxies, logit	9%	8.46	0.964 $\pm$0.001	0.963 $\pm$0.001	0.957	95.1	94.3	94.2	1.00
Galaxies, pow3	13%	1.00	0.945 $\pm$0.001	0.942 $\pm$0.001	0.937	95.1	95.2	96.0	1.00
AlphaFold, logit	23%	9.50	0.900 $\pm$0.001	0.899 $\pm$0.001	0.892	94.7	95.0	95.0	1.00
Forest, logit	5%	4.94	0.975 $\pm$0.001	0.977 $\pm$0.001	0.979	94.7	95.1	95.2	1.00
Gene expression, exp	60%	4.62	0.738 $\pm$0.001	0.731 $\pm$0.001	0.731	94.8	95.4	95.3	1.00
Census health ins., logit	8%	6.50	0.971 $\pm$0.001	0.967 $\pm$0.001	0.962	94.4	94.3	94.5	1.00
Census income, sqrt	8%	1.20	0.987 $\pm$0.002	0.976 $\pm$0.001	0.961	94.6	95.5	94.9	1.20

5.3 Distorted predictor outputs: where recalibration pays

Classifiers often emit scores that are monotone but not affine in the probability of the label (logits, over-confident or squashed scores Niculescu-Mizil and Caruana, 2005, Guo et al., 2017). To measure what that costs, we apply a fixed monotone map to the shipped predictor outputs (logit of the probability score; cubed score; $\exp(\hat{y}/4)$ for Gene expression; $\sqrt{\cdot}$ for Census income) and keep everything else. These are *constructed* cases, chosen to represent the miscalibration mechanism, not evidence of how common it is. Table 2 shows that the width reduction of auto/iso relative to PPI++ ranges from -2% to -27%, matching the bound within about 2 percentage points ($n = 1000$), with nominal coverage; plain PPI (λ=1) becomes 4.9–9.5 times wider than PPI++ for logit scores (and 4.6 times for the exponential distortion) because the scores are no longer on the scale of Y. With logit-scale synthetic scores of moderate sharpness ($s = 0.06$) the gain is only about 2%, with sharper logits ($s = 0.23$) it is 10% at $n = 500$ (bound 11.7%) and 3%, 8%, 10% at $n = 50, 100, 200$ (Appendix Table 5).

5.4 Binary predictions, distribution shift, and a fixed-sets application

Binary predictor. The plankton predictions are 0/1. Then g is affine and Cor. 1(ii) applies: auto/iso/lin are within 0.1% of PPI++ in width for $n \geq 500$ and 1.2% narrower at $n = 100$ (coverage in Table 3, top). The same holds for the ballots application below.

Distribution shift. The labeled plankton records are from 2013 and the unlabeled ones from 2014; the target is the 2014 mean. PPI's rectifier $Y - \hat{Y}$ is approximately stable across years, so PPI (λ=1) keeps 94% coverage until $n = 2000$ and degrades at $n = 10{,}000$ (91%), whereas every interval that estimates the calibration curve on 2013 labels (PPI++ with $\hat{\lambda} < 1$, lin, iso, auto) collapses: auto covers 84% at $n = 500$, 9% at $n = 2000$, 0% at $n = 10{,}000$. Efficiency comes from leaning harder on the stationarity of $\mathbb{E}[Y|f]$, which is exactly what fails under shift. Estimating the calibration curve thus makes the interval *less* robust than the PPI default; we have no remedy and make no claim about shift-robust variants.

Fixed-sets application. For the ballots data ($n = 1027$, $N = 78{,}150$, binary predictions, no ground truth for the unlabeled records) the classical interval has width 0.0591, PPI 0.0133, PPI++ 0.0131 and all cross-fitted intervals 0.0132 (Appendix Table 6); with a binary predictor nothing below PPI++ is possible, in line with the theory. Here $\hat{Y} = 1 \Rightarrow Y = 1$ for all labeled records, so the predictor is nearly perfect on one side.

5.5 Side information beyond the predictor

For the three census data sets the features X are available. Using X in the calibration (effectively a new predictor trained on the labels) is outside the fixed-predictor constraint, but the bound shows how much it could give. Cross-validated gradient-boosting fits of Y on (f, X) lower the residual variance by at most 0.9% (Census income; 0.1% and 0.3% for the other two) relative to a fit on f alone (Appendix Table 7), since f already summarizes X (age/sex, income, etc.); there is no headroom

Table 3: Coverage (%) on plankton (1000 replicates, 95% nominal). Top: labeled and unlabeled data from the same year (2013); bottom: labeled 2013, unlabeled 2014, target is the 2014 mean. $N = 20{,}000$.

setting	n	classical	PPI	PPI++	lin	auto
no shift (2013 only)	100	92.3	95.2	91.7	95.6	95.2
	500	94.3	95.7	94.2	94.8	94.7
	2000	94.8	94.5	94.0	94.3	94.5
	10000	94.0	95.1	94.6	94.7	94.7
shift 2013→2014	100	53.7	94.2	94.1	96.5	96.3
	500	0.4	94.8	94.8	84.2	84.4
	2000	0.0	94.2	94.2	9.1	8.9
	10000	0.0	91.2	0.5	0.0	0.0

Table 4: Coverage over all cells with $n \geq 100$ (real, distorted, synthetic and plankton no-shift; shift cells excluded).

method	cells	mean cov. (%)	min cov. (%)	#cells sig. below 95	#cells sig. above 95
classical	157	94.64	90.9	12	1
PPI	157	94.56	88.5	22	0
PPI++ (ppi_py)	157	94.51	87.7	24	2
PPI++ (unclipped)	157	94.46	87.1	26	2
cross-fit linear	157	94.80	88.9	12	2
cross-fit isotonic	157	94.78	89.0	14	7
cross-fit 10 bins	157	94.81	88.6	14	3
cross-fit spline	157	94.77	88.4	13	2
cross-fit auto	157	94.80	89.1	14	3

here either. We did not run interval-level experiments for X-augmented calibration; the statement is about the population risk only.

5.6 Validity summary

Across all real and synthetic cells with $n \geq 100$ (plankton no-shift cells included, shift cells excluded) Table 4 shows that no method is systematically invalid; coverage is significantly below 95% (Wilson upper limit $< 95\%$) in 12 of 157 cells for classical and the cross-fitted lin, 14 for iso/bin10/auto, but 22–26 for PPI, PPI++ (both versions); the worst cells ($\approx$ 87–91%) are Census income at $n = 100$ (all methods). Because the replicates are shared across methods these counts are strongly correlated, and part of the flags are chance (about 4 of 157 expected under exact 95% coverage), so they should be read as "the cross-fitted intervals are at least as well calibrated as the PPI family", not as exact validity. Cells above 95% are rare (1–7).

6 Discussion and limitations

Answer to the question. With labels, unlabeled data and predictor fixed, and the predictor output as the only information, an interval can be shorter than PPI++ only to the extent that $\mathbb{E}[Y \mid f]$ is not affine, by at most $1 - \sqrt{1 - sN/(n+N)}$ in width (Cor. 1), and only if n is large enough to learn the curve. On the seven data sets that ship with `ppi-python` the unmodified predictors are close to calibrated ($s \leq 2.5\%$), so the available gain is $\leq 1.2\%$ in width and cross-fitted recalibration did not realise it at $n \geq 100$ (it cost 0.5–2% at $n = 500$ and up to 19% at $n = 50$). For predictors whose scores are monotone but not on the scale of the label the gain is real (2–27% here) and is predicted accurately by the bound; there the improvement over PPI ($\lambda = 1$) is much larger still. A hard 0/1 predictor leaves no room. Recalibration also removes the dependence on the scale of the scores,

which PPI/PPI++ do not have. A practical rule that follows from the theory: plot the labeled (f, Y) calibration curve, estimate s and the bound by cross-validation, and recalibrate only if $(N/m)\, s$ is of the order of the estimation cost ($\approx$ a few percent at $n \approx 500$); and use the unclipped PPI++ weight, which costs nothing and avoided losses of 2–6% on two data sets.

Limitations. (1) *Target.* Only the population mean was studied; the closed-form gap uses the mean's influence function. For quantiles or regression coefficients the efficient estimators involve other conditional functions, and our numerical findings do not transfer without new checks. (2) *Information set.* The bound is for estimators that use only the scalar predictor output. Features X could help in principle; we only measured population-level gains with gradient boosting on the three census sets (negligible), not interval-level behaviour, and learning from X is closer to training a new predictor than to the fixed-predictor constraint of the task. (3) *Labeling mechanism.* Labeled points are an i.i.d. random subset (MCAR). Under shift (plankton) the recalibrated intervals were much less robust than PPI; we did not study covariate-shift corrections. (4) *Asymptotics and small n.* Intervals are Wald-type; Prop. 3 is asymptotic and not proved for the CV-selected learner (**auto**). Coverage at $n \leq 100$ is below nominal for all methods on heavy-tailed outcomes (Census income: 82–92%) and can be 1–4 points low even on benign data; we do not claim to fix that, and "shorter" claims are only meaningful where coverage is nominal. A t-quantile did not repair PPI++ at $n = 50$. (5) *Resampled populations.* Real-data cells resample the empirical distribution of finite example sets; for Forest (1596 records) and for n comparable to P, repeated records make the effective problem easier for nonparametric fits. The distortions in Sec. 5.3 are constructed, so we have no estimate of how often real deployed predictors are non-affinely calibrated; the two data sets with the largest s (Gene expression 2.5%, Census income 0.5%) are those whose predictors are regressions on continuous outcomes, which is suggestive but not a sample of predictors. (6) *Novelty.* The efficiency bound and the cross-fitted augmented estimator are classical results in semiparametric/semi-supervised statistics; our contributions are the closed-form comparison with PPI and PPI++, the validity statement without rates, and the empirical map of when the gap is nonzero. Related prior work on the same estimand may exist that we did not find. (7) *Scope of evidence.* 2000 (synthetic) or 1000 (real) replicates per cell give coverage standard errors of about 0.5–0.7 points, so differences below one point in coverage are not resolved; width ratios are paired and much more precise. **Broader impact.** The work is methodological, uses public data and has no foreseeable direct negative societal impact; the main risk is overconfidence from intervals used outside their assumptions (labeling mechanism, shift), which the failure cases document.

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A Proofs

Proposition 1. Since $\mathbb{E}[Y - g(Z) \mid Z] = 0$, $\mathrm{Var}(Y - \lambda Z) = \sigma^2 + \mathrm{Var}(g(Z) - \lambda Z)$. Decompose $g(Z) = \mathbb{E}g + \beta(Z - \mathbb{E}Z) + h(Z)$ with $h \perp \{1, Z\}$ in L^2 (by the definition of β), so $\mathrm{Var}(g - \lambda Z) = v_h + (\beta - \lambda)^2 V$. Hence $V_\lambda = \frac{\sigma^2 + v_h}{n} + V\big[\frac{(\beta-\lambda)^2}{n} + \frac{\lambda^2}{N}\big]$. The bracket is a quadratic in λ with leading coefficient $\frac{1}{n} + \frac{1}{N} = \frac{m}{nN}$, minimized at $\lambda^\star = \beta N/m$, where $\frac{(\beta-\lambda^\star)^2}{n} + \frac{\lambda^{\star 2}}{N} = \beta^2\big(\frac{n}{m^2} + \frac{N}{m^2}\big) = \frac{\beta^2}{m}$. The formula for $V_\lambda - V_{\mathrm{PPI}++}$ is the quadratic's expansion around its minimizer.

Corollary 1. $V_{\mathrm{PPI}++} - V_{\mathrm{eff}} = \frac{v_h}{n} - \frac{v_h}{m} = v_h \frac{N}{nm}$. Dividing by $V_{\mathrm{PPI}++} = \frac{(\sigma^2 + v_h)(1+\kappa)}{n}$ gives the ratio. For a two-point Z every function of Z is affine, so $h \equiv 0$.

Proposition 3. Let $\tilde{\theta} = \frac{1}{n}\sum_L (Y_i - g^*(Z_i)) + \frac{1}{m}\sum_{i \le m} g^*(Z_i)$ be the oracle estimator. Because L is a uniformly random subset of an i.i.d. sample, $\tilde{\theta} - \theta$ is a sum of two uncorrelated terms (with $W_i = Y_i - g^*(Z_i)$, $\tilde{\theta} - \theta = \bar{W}_L - \mathbb{E}W + \ldots$); its variance is $\mathbb{E}(Y - g^*)^2/n + \mathrm{Var}\, g^*(Z)/m$ up to $O(1/m)$-relative terms and a Lindeberg-type CLT for sampling without replacement from an i.i.d. sample applies. Write $\delta_k = \hat{g}_{-k} - g^*$. Then

$$\hat{\theta} - \tilde{\theta} = \sum_{k=1}^{K} \Big[\Big(\tfrac{1}{m} - \tfrac{1}{n}\Big) \sum_{i \in L_k} \delta_k(Z_i) + \tfrac{1}{Km} \sum_{j \notin L} \delta_k(Z_j)\Big] =: \sum_k D_k.$$

Conditional on the training data for fold k, the points $Z_i, i \in L_k$ ($|L_k| = n/K$) and $Z_j, j \notin L$ (N points) are i.i.d. and independent of δ_k, so $\mathbb{E}[D_k \mid \text{train}] = \mathbb{E}\delta_k(Z)\big[(\frac{1}{m} - \frac{1}{n})\frac{n}{K} + \frac{N}{Km}\big] = 0$ (exact unbiasedness for any $\hat{g}$) and $\mathrm{Var}(D_k \mid \text{train}) \le C\,\|\delta_k\|_{L^2}^2/n$ with C depending only on n/m. By conditional Chebyshev, $D_k = o_p(n^{-1/2})$ when $\|\delta_k\|_{L^2} \to_p 0$; no rate is needed. Hence $\hat{\theta} - \theta = (\tilde{\theta} - \theta) + o_p(n^{-1/2})$. The same argument applied to the sample variances (using

$\widehat{\mathrm{Var}}_L(Y-\hat{g}) = \widehat{\mathrm{Var}}_L(Y-g^*) + o_p(1)$, which follows from $\|\delta_k\|_{L^2} \to_p 0$ and Cauchy–Schwarz, and similarly for $\widehat{\mathrm{Var}}_{\mathrm{all}}(\hat{g}))$ shows $\hat{V}/V^* \to_p 1$. Slutsky's lemma finishes the proof. If $g^* = g$, $V^* = V_{\mathrm{eff}}$ by Proposition 2. □

Remark (what is *not* proved). The proposition requires $\hat{g}_{-k}$ to converge to a deterministic limit. For the data-driven choice among lin/iso/spl this holds when the candidates' limits have different risks, but not necessarily when two of them are tied with different limits; we do not prove coverage of **auto** and evaluate it by simulation. All results are asymptotic in n and N; no finite-sample guarantee is claimed.

B Additional results

Table 5: Logit-scale synthetic classifier ($Y \sim \mathrm{Bernoulli}(\mathrm{expit}(f))$, $f \sim N(-1, \tau^2)$; top $\tau = 2$, bottom $\tau = 4$); 2000 replicates. Widths relative to PPI++ (unclipped); bound from the closed form with exact moments.

logit scores	n	N	width / PPI++ (unclipped)				coverage (%)		
			PPI	auto	iso	bound	PPI++	iso	auto
s=0.06	50	20000	4.58	1.055 $\pm$0.005	1.041 $\pm$0.003	0.970	93.3	93.8	94.2
	200	20000	4.58	0.996 $\pm$0.001	0.997 $\pm$0.001	0.971	94.8	94.7	94.5
	500	500	6.14	1.001 $\pm$0.000	1.011 $\pm$0.000	0.988	94.8	95.2	95.0
	500	5000	4.73	0.984 $\pm$0.000	0.990 $\pm$0.001	0.974	95.8	96.0	95.6
	500	20000	4.60	0.981 $\pm$0.000	0.986 $\pm$0.001	0.971	95.5	96.0	95.5
	500	50000	4.57	0.981 $\pm$0.000	0.985 $\pm$0.001	0.971	94.8	95.0	95.0
	3000	20000	4.82	0.978 $\pm$0.000	0.981 $\pm$0.000	0.976	95.6	95.5	95.6
s=0.23	50	20000	10.65	0.967 $\pm$0.006	0.946 $\pm$0.005	0.877	94.0	94.2	94.0
	200	20000	10.74	0.904 $\pm$0.002	0.905 $\pm$0.002	0.879	95.3	95.2	95.2
	500	500	12.83	0.978 $\pm$0.000	0.983 $\pm$0.000	0.962	94.3	95.0	94.7
	500	5000	10.88	0.914 $\pm$0.001	0.914 $\pm$0.001	0.899	94.3	94.8	94.7
	500	20000	10.79	0.899 $\pm$0.001	0.898 $\pm$0.001	0.883	95.0	95.2	95.2
	500	50000	10.76	0.895 $\pm$0.001	0.894 $\pm$0.001	0.879	94.5	95.2	95.5
	3000	20000	10.93	0.913 $\pm$0.000	0.913 $\pm$0.000	0.908	94.9	95.1	95.0

Table 6: Ballots application (real fixed sets; no ground truth). Cross-fitted rows average the width over 20 fold-assignment seeds (s.d. over seeds $\leq 2 \times 10^{-5}$).

method	lower	upper	width
classical	0.5985	0.6576	0.05912
PPI	0.6145	0.6278	0.01329
PPI++ (ppi_py)	0.6149	0.6280	0.01312
PPI++ (unclipped)	0.6149	0.6280	0.01312
cross-fit linear	0.6149	0.6280	0.01315
cross-fit isotonic	0.6149	0.6280	0.01315
cross-fit 10 bins	0.6149	0.6280	0.01315
cross-fit spline	0.6149	0.6280	0.01315
cross-fit auto	0.6149	0.6280	0.01315

Table 7: Residual variance of Y (relative to $\mathrm{Var}(Y)$) from 5-fold cross-validated fits on the census sets: linear in f, gradient boosting (GB) on f, on X, on (f, X). The last column is the relative reduction from adding X to f.

data set	lin(f)	GB(f)	GB(X)	GB(f, X)	gain of X over GB(f)
Census income	0.6309	0.6283	0.8474	0.6254	0.47%
Census health ins.	0.5004	0.5005	0.8428	0.5000	0.09%
Census education	0.6079	0.6080	0.8457	0.6062	0.29%

Table 8: Real data, shipped predictors, $N = 20{,}000$, varying n (1000 replicates). Width ratios are relative to PPI++ (unclipped).

	n=50	n=100	n=200	n=500	n=1000	n=2000	n=5000
width of auto / width of PPI++ (unclipped)							
Galaxies	1.058	1.031	1.016	1.007	1.004	1.002	1.001
Forest	1.146	1.032	1.014	1.005	0.999	0.994	0.991
AlphaFold	1.027	1.014	1.011	1.005	1.003	1.002	1.000
Gene expression	0.976	0.987	0.992	0.994	0.994	0.996	0.998
Census income	1.191	1.085	1.043	1.021	1.011	1.006	1.002
Census health ins.	1.064	1.035	1.018	1.008	1.004	1.002	1.001
Census education	1.113	1.049	1.027	1.011	1.005	1.003	1.001
width of PPI (λ=1) / width of PPI++ (unclipped)							
Galaxies	0.984	0.992	0.996	1.000	1.002	1.007	1.030
Forest	0.984	0.994	0.999	1.003	1.005	1.009	1.023
AlphaFold	0.943	0.971	0.985	0.994	0.998	1.002	1.024
Gene expression	1.017	1.055	1.060	1.049	1.027	1.004	1.010
Census income	1.012	1.019	1.019	1.018	1.015	1.010	1.002
Census health ins.	0.996	0.999	0.999	1.000	1.002	1.006	1.024
Census education	1.000	1.000	1.001	1.001	1.002	1.004	1.018
coverage (%) of auto							
Galaxies	94.2	95.3	95.4	95.2	95.1	95.3	94.9
Forest	92.3	94.0	94.5	95.1	94.6	95.2	95.9
AlphaFold	89.2	93.3	94.3	94.2	94.3	95.1	94.2
Gene expression	95.6	94.7	94.3	93.8	95.2	95.3	94.0
Census income	84.8	89.3	91.7	93.4	94.2	94.8	95.3
Census health ins.	94.7	94.5	95.8	95.0	95.0	96.3	94.2
Census education	93.6	93.1	93.3	94.5	94.8	94.5	94.7
coverage (%) of PPI++ (unclipped)							
Galaxies	92.8	93.7	95.1	94.7	94.7	95.3	94.7
Forest	89.5	92.9	93.5	95.1	94.9	95.1	95.9
AlphaFold	86.6	92.3	91.9	94.3	94.0	95.0	94.1
Gene expression	93.2	93.2	93.8	94.0	95.2	95.2	94.5
Census income	81.6	87.1	90.4	93.1	93.4	94.7	95.5
Census health ins.	94.3	93.9	95.6	94.8	95.1	96.1	94.3
Census education	91.7	92.4	93.0	94.1	94.8	94.7	94.6

C Reproducibility details

Software and data. Python 3.11.2, numpy 2.4.6, scipy 1.17.1, pandas 3.1.1, scikit-learn 1.9.1, matplotlib 3.11.2, `ppi-python` 0.2.3 (MIT; the `.npz` example data carry no separate license statement in the repository; original sources are cited in its example notebooks), gdown 6.5.1. The data are downloaded by `ppi_py.datasets.load_dataset`. **Seeds.** Replicate r of cell c uses `SeedSequence([20261111, crc32(c), r])`. **Compute.** CPU only; the container exposes 4 effective cores (cgroup quota) on a shared host; the full set of experiments (E1a–E4) took about 2.5 wall-clock hours, i.e. roughly 10 CPU-hours. Per-experiment commands and the mapping from every table and figure to its script are in the supplementary `README.md`. Preliminary runs (a 40-replicate smoke test of the runner and an aborted first launch that was restarted after replacing a slow spline

Table 9: Real data, shipped predictors, $n = 500$, varying the number of unlabeled points N (1000 replicates).

	N=50	N=250	N=1000	N=5000	N=50000
width of auto / width of PPI++ (unclipped), $n = 500$					
Galaxies	1.009	1.009	1.008	1.007	1.007
Forest	1.019	1.016	1.011	1.006	1.004
AlphaFold	1.011	1.011	1.010	1.006	1.003
Gene expression	1.003	1.003	1.002	0.998	0.991
Census income	1.025	1.024	1.021	1.021	1.021
Census health ins.	1.009	1.009	1.008	1.008	1.007
Census education	1.014	1.014	1.012	1.011	1.011
width of PPI (λ=1*) / width of PPI++ (unclipped)*					
Galaxies	2.507	1.402	1.079	1.007	0.999
Forest	2.060	1.264	1.053	1.008	1.001
AlphaFold	2.582	1.423	1.079	0.999	0.994
Gene expression	2.753	1.468	1.072	1.002	1.073
Census income	1.662	1.102	1.005	1.010	1.020
Census health ins.	2.421	1.355	1.066	1.005	1.000
Census education	2.200	1.287	1.052	1.005	1.000
width of classical / width of PPI++ (unclipped)					
Galaxies	1.027	1.106	1.259	1.416	1.482
Forest	1.014	1.053	1.115	1.165	1.186
AlphaFold	1.033	1.135	1.338	1.590	1.703
Gene expression	1.046	1.204	1.619	2.519	3.480
Census income	1.018	1.070	1.158	1.238	1.269
Census health ins.	1.024	1.096	1.225	1.357	1.407
Census education	1.019	1.073	1.165	1.249	1.279
coverage (%) of auto					
Galaxies	94.9	94.8	96.0	96.0	95.2
Forest	96.8	94.6	95.7	93.8	94.3
AlphaFold	95.1	95.6	94.9	95.7	95.6
Gene expression	95.9	95.7	96.4	95.7	95.3
Census income	95.8	94.2	93.6	92.8	94.1
Census health ins.	95.6	95.1	95.4	94.6	94.4
Census education	93.9	93.6	94.6	95.7	94.9

basis by a faster natural-spline implementation) are not reported; no other experiments were run and discarded.

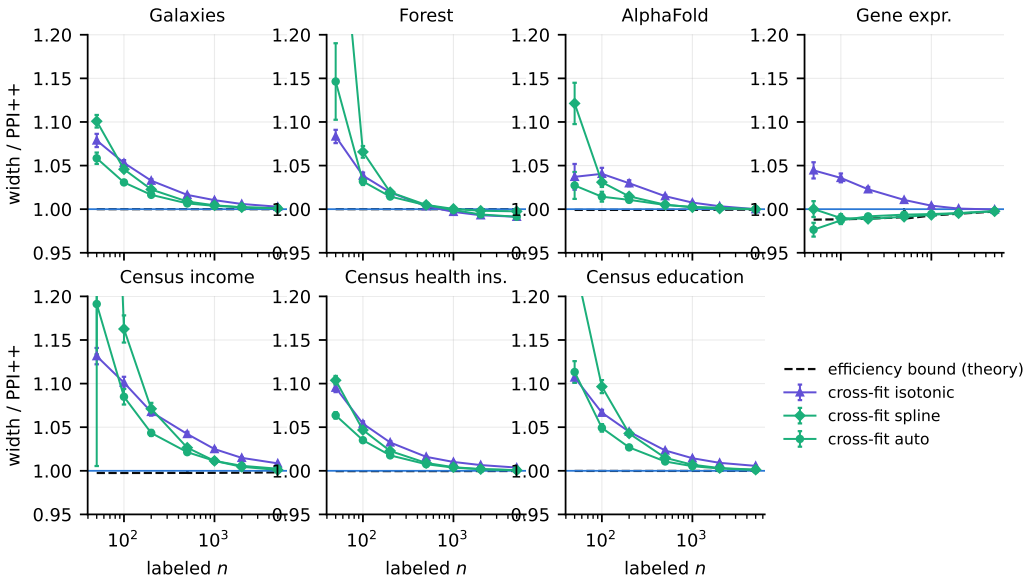


Figure 3: Real data, shipped predictors: width of cross-fitted intervals relative to PPI++ (unclipped) versus n ($N = 20{,}000$; ± 1.96 s.e.). Dashed: bound estimated from the population. Classical and PPI curves are off this scale (Table 8). The Forest population has 1596 records, so $n \geq 1000$ is affected by repeated records.

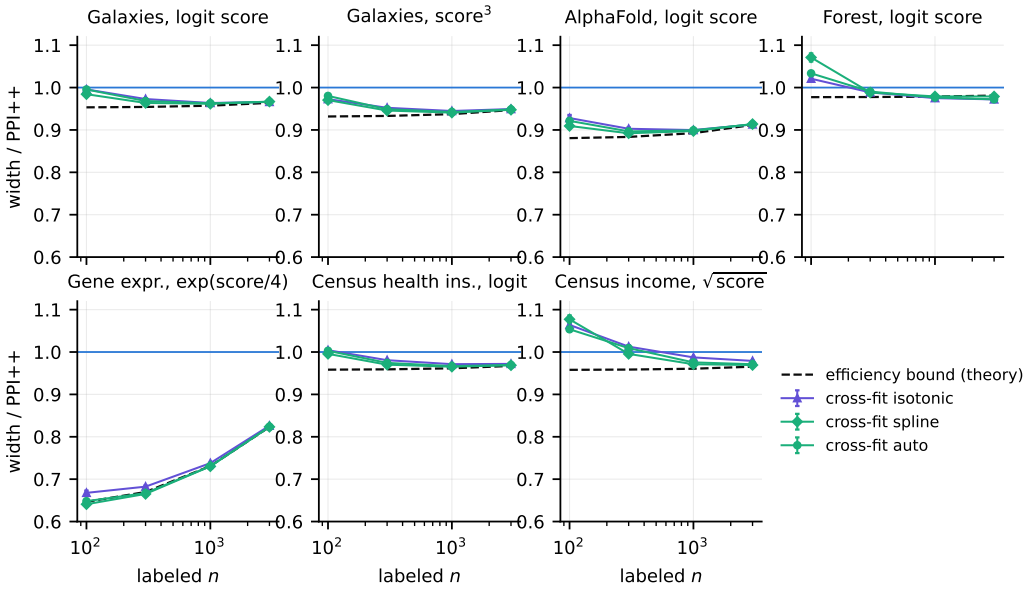


Figure 4: Distorted predictor outputs: same quantities as above ($N = 20{,}000$).

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